# IGCSE 0607 Practice Questions — E4.5 — Combined Questions & Markscheme

## Parallel lines

**Total Questions:** 6 — **Total Marks:** 29 Marks

## Table of Contents & Quick Navigation


- **E4.5** Parallel lines
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

**1** Line $L$ has the equation $y = 6 - 3x$.

Find the equation of the line parallel to $L$ that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**2** The points $A$ and $B$ have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line $L$ is parallel to the line segment $AB$ and passes through the point $(4, -2)$.

**(a)** Find the gradient of the line segment $AB$.

..... **[2]**

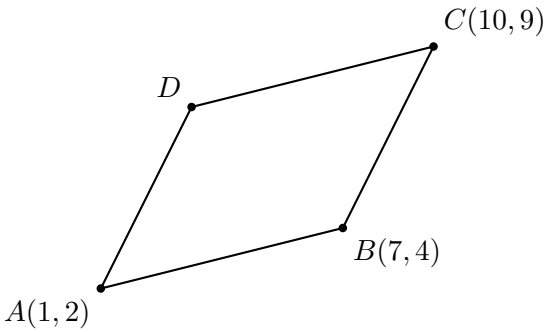
**(b)** Find the equation of line $L$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**3** Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**4** The points $A(1, 2)$, $B(7, 4)$, and $C(10, 9)$ are three vertices of a parallelogram $ABCD$.
Line $CD$ is parallel to line $AB$ and line $AD$ is parallel to line $BC$.




*Not to scale*

**(a)** Find the equation of the line $CD$.

..... **[3]**

**(b)** Find the equation of the line $AD$.

..... **[3]**

**(c)** Work out the coordinates of the vertex $D$.

( ..... , ..... ) **[3]**

**5** Line $M$ has the equation $5x - 2y = 15$.
Point $P$ has coordinates $(-4, 1)$.
Find the equation of the line parallel to $M$ that passes through $P$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**6** Points $A(-1, 5)$ and $B(3, -3)$ lie on line $L_1$.
Line $L_2$ is parallel to $L_1$ and passes through the point $C(2, 7)$.
Line $L_2$ crosses the $x$-axis at $D$ and the $y$-axis at $E$.

**(a)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**(b)** Calculate the exact area of triangle $ODE$, where $O$ is the origin $(0, 0)$.

..... $units^2$ **[3]**

**1** Gradient $m = -3$ **[B1]**

$5 = -3(-2) + c$ or $y - 5 = -3(x - (-2))$ **[M1]**

$y = -3x - 1$ **[A1]**

[Back to Question] *[Total Marks: 3]*

**2 (a)** $\frac{-1-5}{5-(-3)}$ or $\frac{5-(-1)}{-3-5}$ **[M1]**

$-\frac{3}{4}$ **[A1]**

**(b)** $m = -\frac{3}{4}$ **[M1]**

$-2 = -\frac{3}{4}(4) + c$ or $y - (-2) = -\frac{3}{4}(x - 4)$ **[M1]**

$y = -\frac{3}{4}x + 1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**3** Identifies the gradient $m = -2$ and sets up $4 = -2(3) + c$ or $y - 4 = -2(x - 3)$ **[M1]**

$-2x + 10$ **[A1]**

[Back to Question] *[Total Marks: 2]*

**4 (a)** Gradient of $AB = \frac{4-2}{7-1} = \frac{1}{3}$ **[M1]**

Substitute $(10, 9)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 9 = \frac{1}{3}(x - 10)$ **[M1]**

$y = \frac{1}{3}x + \frac{17}{3}$ or $3y = x + 17$ **[A1]**

**(b)** Gradient of $BC = \frac{9-4}{10-7} = \frac{5}{3}$ **[M1]**

Substitute $(1, 2)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 2 = \frac{5}{3}(x - 1)$ **[M1]**

$y = \frac{5}{3}x + \frac{1}{3}$ or $3y = 5x + 1$ **[A1]**

**(c)** Equate equations from (a) and (b): $\frac{1}{3}x + \frac{17}{3} = \frac{5}{3}x + \frac{1}{3}$ **[M1]**

$4x = 16 \implies x = 4$ **[A1]**

$y = 7$ **[A1]**

[Back to Question] *[Total Marks: 9]*

**5** Rearranges $5x - 2y = 15$ to find the gradient, e.g. $2y = 5x - 15$

y $= \frac{5}{2}x - 7.5$, so $m = \frac{5}{2}$ [M1]

Substitutes $(-4, 1)$ and $m = \frac{5}{2}$ into $y = mx + c$: $1 = \frac{5}{2}(-4) + c$ **[M1]**

$\frac{5}{2}x + 11$ or $2.5x + 11$ **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**6** **(a)** Calculates the gradient of $L_1$: $m = \frac{-3-5}{3-(-1)}$ **[M1]**

$m = \frac{-8}{4} = -2$ **[A1]**

Uses parallel gradient $-2$ and point $C(2, 7)$ to find $c$: $7 = -2(2) + c$ **[M1]**

$-2x + 11$ **[A1]**

**(b)** Finds $y$-intercept $E(0, 11)$ and sets $y = 0$ to find $x$-intercept $D$: $0 = -2x + 11$

x $= 5.5[M1]$

Sets up area calculation for right-angled triangle: $\frac{1}{2} \times 5.5 \times 11$ **[M1]**

30.25 or $\frac{121}{4}$ **[A1]**

**[Back to Question]** *[Total Marks: 7]*